Problem Set 2: Viral entry

Physics 230A/146A — Biological Physics

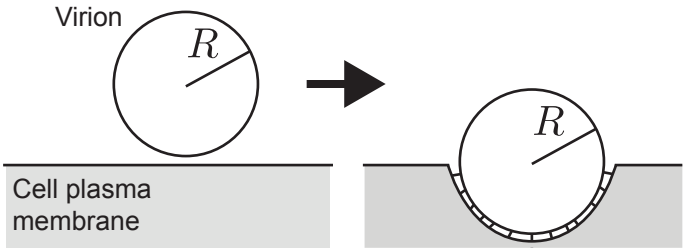
Jun Allard

Membranes, including the plasma membrane that constitutes the outer envelope of eukaryotic cells, are two-dimensional sheets composed of lipid bilayers that resist mechanical deformation, like bending. The bending energy of a surface can be modeled as

$$E = \iint \frac{1}{2} B\kappa^2 \, dA$$

where $B = 100\,\text{pN nm}$ is the bending modulus (note it has units of pN nm, unlike for rods and filaments), and κ is the mean curvature, and the integral $\iint dA$ is over the membrane's surface. For a sphere, $\kappa = 2/R$ where R is the radius of the sphere. For a cylinder, $\kappa = 1/R$ where R is the radius of the cylinder.

Viruses spread between cells as virus particles, or virions. In this question, we will explore the energetic constraints on virion entry into the host cell. Can it happen passively, or does it require active (energy-consuming) machinery?



i. Suppose a viral capsid is approximately spherical with radius $R = 200\,\text{nm}$. How much energy is required to bend the plasma membrane to get the capsid half-way into the host cell?

ii. Suppose the cell is under turgor pressure of $P = 0.01\,\text{pN/nm}^2$. Now, locally changing the volume of the cytoplasm results in an increase in energy $\Delta E = PV$, in addition to the energy required to bend the membrane. How much energy (bending plus volume change) is required to get the capsid half-way into the host cell?